

Internship Final Report

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Major: BCA-Data analytics

Internship Duration: July 1st, 2025 – July 31st, 2025

Company: ShadowFox

Domain: Data Science

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

OBJECTIVE

The primary objective of the beginner-level task during the ShadowFox Data Science Internship was to create a comprehensive documentation guide for two popular Python visualization libraries. The purpose was to help beginners understand the various types of visualizations available in each library, along with their use cases and implementation through code snippets. This task aimed to build our technical writing and analytical skills while also enhancing our knowledge of data visualization tools commonly used in the data science industry.

TASKS AND RESPONSIBILITIES

As part of the beginner-level task, I selected **Matplotlib** and **Seaborn** as my two visualization libraries. My responsibilities included:

- Studying and understanding the functionalities of both libraries using their official documentation.
- Writing an overview for each library, covering its features and common use cases.
- Documenting a variety of graph types such as line plots, bar charts, scatter plots, histograms, and pie charts.
- Providing use cases and sample Python code for each graph type to demonstrate practical implementation.
- Creating a comparison table outlining the differences between Matplotlib and Seaborn in terms of ease of use, customization, interactivity, and performance.
- Compiling all the findings into a clear and concise PDF report and uploading the source code to my GitHub repository named "ShadowFox."
- Creating a short video explanation and sharing it on LinkedIn as proof of work.

LEARNING OUTCOMES

This task allowed me to:

- Gain hands-on experience with two major Python visualization libraries.
 - Learn how to communicate technical content effectively through documentation.
 - Understand the strengths and limitations of Matplotlib and Seaborn in real-world data visualization.
 - Improve my coding skills by working on clean, functional examples using Python.
 - Become familiar with GitHub repositories, documentation structure, and sharing technical projects online.
 - Develop confidence in presenting my work to a public audience via LinkedIn video.
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CHALLENGES AND SOLUTIONS

Challenge 1: Understanding advanced customizations in Matplotlib plots.

Solution: I referred to the official documentation and experimented with different parameters to understand how plot appearance can be adjusted.

Challenge 2: Formatting the content in a beginner-friendly and structured way.

Solution: I created a layout that separates the overview, graph types, and code examples clearly. I also used bullet points and tables for better readability.

Challenge 3: Managing GitHub uploads and version control.

Solution: I used GitHub Desktop for smoother workflow and watched tutorials to understand push/pull operations effectively.

CONCLUSION

The beginner-level task in the ShadowFox Data Science Internship was a valuable experience that strengthened both my technical and professional skills. It enabled me to explore the world of data visualization and understand how to make data more understandable and presentable. I gained a solid foundation in Matplotlib and Seaborn, which I will continue to use in future projects. This task was not only educational but also boosted my confidence in documenting and sharing my work publicly.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **ShadowFox** for providing me with this incredible opportunity to participate in the Data Science Internship. This program has significantly contributed to my technical growth and helped me understand real-world applications of data science.

I am especially thankful to **Mr. Hariharan**, our mentor, whose guidance and insights were extremely valuable throughout the beginner-level task. His mentorship helped me better understand the tools and techniques required to approach technical documentation effectively.

I would also like to thank **Mr. Aakash**, the program coordinator, for his continuous support, timely updates, and responsiveness to queries during the internship. His structured communication made it easy to stay on track with deadlines and task submissions.

Additionally, I appreciate the efforts of the entire ShadowFox team for designing such a beginner-friendly, practical internship experience. The resources, task formats, and Proof of Work model provided a real sense of industry exposure.

This internship not only strengthened my skills but also boosted my confidence in working independently and presenting my work professionally.

